

Experiment-5

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Course Code: -MLA0305

Slot-B

Title: -Dynamic Programming Using Policy Iteration and Value Iteration

Aim-To implement Policy Iteration and Value Iteration using Dynamic Programming and determine the optimal policy and value function for a Reinforcement Learning environment.

Procedure

A. Policy Iteration

1. Define the environment and initialize a random policy.
2. Initialize the value function.
3. Perform policy evaluation using the Bellman equation.
4. Calculate the value of each state under the current policy.
5. Perform policy improvement.
6. Select the action with the maximum expected value.
7. Update the policy.
8. Repeat policy evaluation and policy improvement until the policy becomes stable.

B. Value Iteration

9. Initialize the value function.

$$V(s) = \max_a \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V(s')]$$

11. Update the value function repeatedly.
12. Stop when the values converge.
13. Extract the optimal policy from the final value function.
14. Compare the results of Policy Iteration and Value Iteration.

Source Code

```
import numpy as np
import pandas as pd import matplotlib.pyplot
as plt from sklearn.mixture import
GaussianMixture from IPython.display import
display, FileLink from google.colab import
files np.random.seed(42) print("=" * 80)
print("EXPERIMENT 5")
print("DYNAMIC PROGRAMMING USING POLICY ITERATION AND VALUE ITERATION")
print("=" * 80) N = 500 base = np.column_stack([
np.random.uniform(0, 15, N), np.random.uniform(0, 15, N),
np.random.uniform(-5, 10, N),
```

```

        np.random.uniform(0.85, 0.99, N)
    ]) gmm =
    GaussianMixture(
        n_components=3,
        random_state=42
    )
    gmm.fit(base)
    generated =
    gmm.sample(N)[0]
    dataset =
    pd.DataFrame(
        generated,
        columns=[
            "State",
            "Next_State",
            "Reward",
            "Gamma"
        ]
    ) dataset["State"] = dataset["State"].clip(0,
15).round().astype(int) dataset["Next_State"] =
dataset["Next_State"].clip(0,
15).round().astype(int) dataset["Reward"] =
dataset["Reward"].clip(-5, 10).round(2) dataset["Gamma"] =
dataset["Gamma"].clip(0.85, 0.99).round(3)

# ===== #
2. 10 SAMPLE RECORDS
# =====

print("\n10 SAMPLE DATASET RECORDS")
    display(dataset.head(10))
    print("Total Records:",
len(dataset))

# ===== #
3. DATASET SUMMARY
# =====

    display(
dataset.describe().T[
["min", "max", "mean",
"std"]
    ].round(3)
)

```

```

# ===== #
4. DOWNLOAD
# =====

filename =
"Experiment_5_Dynamic_Programming_GMM_Dataset.csv"
dataset.to_csv(
filename,
index=False
) print("\nDOWNLOAD
DATASET:")
display(FileLink(filename))
files.download(filename)

# ===== #
5. GRIDWORLD
# =====

ROWS = 4
COLS = 4
states = [      (r, c)
for r in range(ROWS)
for c in range(COLS)
]

GOAL = (3, 3)

OBSTACLES = [
    (1, 1),
    (2, 2)
]

ACTIONS = {
    "UP": (-1, 0),
    "DOWN": (1, 0),
    "LEFT": (0, -1),
    "RIGHT": (0, 1)
}

ARROWS = {
    "UP": "↑",
    "DOWN": "↓",
    "LEFT": "←",
    "RIGHT": "→"
}

```

```

GAMMA = 0.90

# ===== #
6. TRANSITION
# =====

def transition(state,
action):
    if state == GOAL:
return state, 0
    dr, dc =
ACTIONS[action]
    nr = state[0] +
dr    nc = state[1] +
dc
    if nr < 0 or nr >= ROWS or nc < 0 or nc >=
COLS:
        return state, -2
    ns = (nr,
nc)
    if ns in OBSTACLES:
return state, -5
    if ns == GOAL:
return ns, 10
    return ns, -
1

# ===== #
7. VALUE ITERATION
# =====

V_vi = { s:
    0.0
    for s in states
} for o in
OBSTACLES:
    V_vi[o] = np.nan

V_vi[GOAL] = 10
vi_delta_history =
[]
for vi_iteration in range(1,
501):
    new_V =
V_vi.copy()    delta =
0

```

```

        for s in
states:
    if s in OBSTACLES or s ==
GOAL:
        continue
    values =
[]
        for a in
ACTIONS:
            ns, reward = transition(s,
a)
            q = reward + GAMMA *
V_vi[ns]

values.append(q)
    new_V[s] =
max(values)

    delta = max(
delta,
        abs(new_V[s] -
V_vi[s])
    )

    V_vi = new_V

vi_delta_history.append(delta)
    if delta <
0.001:
        break

# ===== #
8. VALUE ITERATION POLICY
# =====

policy_vi =
{}
    for s in
states:
        if s in
OBSTACLES:
            policy_vi[s] = "BLOCK"
        continue
        if s == GOAL:
policy_vi[s] = "GOAL"
        continue

```

```

        action_values =
    {}
        for a in
ACTIONS:
            ns, reward = transition(s,
a)
            action_values[a] =
(
                reward +
                GAMMA * V_vi[ns]
            )
            policy_vi[s] =
max(
            action_values,
key=action_values.get
        )

# ===== #
9. POLICY ITERATION
# =====

V_pi = {
s:
0.0
for s in
states
}
for o in
OBSTACLES:
    V_pi[o] = np.nan

V_pi[GOAL] = 10
policy_pi = {}
for s in states:
    if s in OBSTACLES or s == GOAL:
        policy_pi[s] = None
    else:
        policy_pi[s] = "RIGHT"
pi_iteration_history =
[]
for pi_iteration in range(1,
101):

    # -----
    # Policy Evaluation
    # -----

    for _ in
range(100):
        new_V =
V_pi.copy()
        for s in
states:

```

```

        if s in OBSTACLES or s ==
GOAL:
            continue
        ns, reward =
transition(
            s,
policy_pi[s]
        )
        new_V[s] = (
reward +
            GAMMA * V_pi[ns]
        )

        V_pi = new_V

# -----
# Policy Improvement
# -----
    stable =
True
    for s in states:
        if s in OBSTACLES
        or s == GOAL:
            continue
        old_action =
policy_pi[s]
        action_values =
        {}
        for a in
ACTIONS:
            ns, reward = transition(s,
a)
            action_values[a] =
            (
                reward +
                GAMMA * V_pi[ns]
            )
            best_action = max(
action_values,
key=action_values.get
            )
            policy_pi[s] =
best_action
            if old_action !=
best_action:
                stable = False

pi_iteration_history.append(pi_iteration)

```

```

        if
stable:
break

# ===== #
10. RESULT COMPARISON
# =====

    valid_states = [      s
for s in states      if s
not in OBSTACLES      and
s != GOAL
]
vi_values = np.array([
    V_vi[s]
    for s in valid_states
])
pi_values = np.array([
V_pi[s]      for s in
valid_states
]) comparison =
pd.DataFrame({
"Algorithm": [
    "Value Iteration",
    "Policy Iteration"
],

    "Iterations": [
vi_iteration,
pi_iteration
],

    "Maximum Value": [
vi_values.max(),
pi_values.max()
],

    "Minimum Value": [
vi_values.min(),
pi_values.min()
],

    "Average Value": [
vi_values.mean(),
pi_values.mean()
]

```



```

}).round(3) print("\n" + "=" * 80)
print("POLICY ITERATION VS VALUE ITERATION")
print("=" * 80)
display(comparison)

# =====
# 11. SUMMARY TABLE
# =====
summary = pd.DataFrame({      "Metric": [
    "Dataset Records",
    "Grid Size",
    "Total States",
    "Obstacles",
    "Goal",
    "Gamma",
    "VI Iterations",
    "PI Iterations",
    "VI Average Value",
    "PI Average Value"
],

    "Result": [
len(dataset),      "4 x 4",
len(states),
len(OBSTACLES),      str(GOAL),
GAMMA,      vi_iteration,
pi_iteration,
round(vi_values.mean(), 3),
round(pi_values.mean(), 3)
    ]
}) print("\nSUMMARY
RESULTS")
display(summary)

# =====
# 12. VISUALIZATION 1
# DATASET REWARD DISTRIBUTION
# =====
plt.figure(figsize=(10,
6))
plt.hist(
dataset["Reward"],
bins=25,
color="#845ef7",
edgecolor="black"
)

```

```

plt.title(
    "GMM Synthetic Dataset - Reward Distribution",
    fontsize=16,      fontweight="bold"
) plt.xlabel("Reward")
plt.ylabel("Frequency")
plt.show()

# =====
# 13. VISUALIZATION 2
# VI CONVERGENCE
# =====

plt.figure(figsize=(11,
6))
plt.plot(
    vi_delta_history,
    color="#e03131",
    linewidth=3,
    marker="o",
    markersize=3
)
plt.title(
    "Value Iteration Convergence",
    fontsize=16,      fontweight="bold"
)
plt.xlabel("Iteration") plt.ylabel("Maximum
Change")
plt.grid(alpha=0.25)
plt.show()

# =====
# 14. VISUALIZATION 3
# ITERATION COMPARISON
# =====

plt.figure(figsize=(9, 6))
plt.bar(
    ["Value Iteration", "Policy Iteration"],
    [vi_iteration, pi_iteration],      color=["#339af0",
"#51cf66"],      edgecolor="black"
)
plt.title(
    "Number of Iterations Comparison",
    fontsize=16,      fontweight="bold"
)
plt.ylabel("Iterations")
plt.show()

```

```

# =====
# 15. VISUALIZATION 4
# VALUE COMPARISON
# =====

plt.figure(figsize=(11,
6))
x =
np.arange(len(valid_states))
plt.plot(x,
vi_values, marker="o",
linewidth=3,
color="#339af0",
label="Value Iteration"
) plt.plot(x,
pi_values, marker="s",
linewidth=3,
color="#ff922b",
label="Policy Iteration"
)
plt.title(
"State Value Comparison",
fontsize=16, fontweight="bold"
) plt.xlabel("State
Index") plt.ylabel("State
Value")
plt.legend()
plt.grid(alpha=0.25)
plt.show()

# =====
# 16. VISUALIZATION 5
# POLICY MAPS
# =====

fig, axes =
plt.subplots(
1, 2,
figsize=(14, 6)
) for ax, policy, title
in [
(ax[0], policy_vi, "Value Iteration Policy"),
(ax[1], policy_pi, "Policy Iteration Policy")
]:
values = np.zeros(
(ROWS, COLS)

```

```

        )
        for r in
range(ROWS):
        for c in
range(COLS):
            if (r, c) in OBSTACLES:
values[r, c] = np.nan
            else:
                values[r, c] = 1
        ax.imshow(
values,
cmap="viridis"
        )
        for r in
range(ROWS):
        for c in
range(COLS):
            state = (r,
c)
            if state in
OBSTACLES:
                text = "■"
            elif state == GOAL:
                text = "★"
            else:
                text = ARROWS[policy[state]]
            ax.text(
c,
r,
text,
ha="center",
va="center",
fontsize=25,
color="white",
fontweight="bold"
            )
        ax.set_title(
title,
fontsize=15,
fontweight="bold"
        )
    plt.tight_layout() plt.show()
    print("\nExperiment 5 completed successfully.")

```

	State	Next_State	Reward	Gamma
0	6	0	-0.89	0.938
1	5	9	-2.71	0.850
2	7	15	-0.81	0.935
3	2	12	-4.97	0.914
4	1	9	-0.90	0.929
5	9	11	-1.54	0.908
6	13	7	-2.69	0.873
7	0	5	-2.06	0.850
8	8	12	-4.05	0.925
9	5	8	0.57	0.895

FIG: - 1 SAMPLE DATA

	Algorithm	Iterations	Maximum Value	Minimum Value	Average Value
0	Value Iteration	7	19.0	7.124	13.202
1	Policy Iteration	5	19.0	7.124	13.202

FIG: - 2 ALGORITHM COMPARISON

SUMMARY RESULTS

	Metric	Result
0	Dataset Records	500
1	Grid Size	4 x 4
2	Total States	16
3	Obstacles	2
4	Goal	(3, 3)
5	Gamma	0.9
6	VI Iterations	7
7	PI Iterations	5
8	VI Average Value	13.202
9	PI Average Value	13.202

FIG: - 3 SUMMARY RESULTS

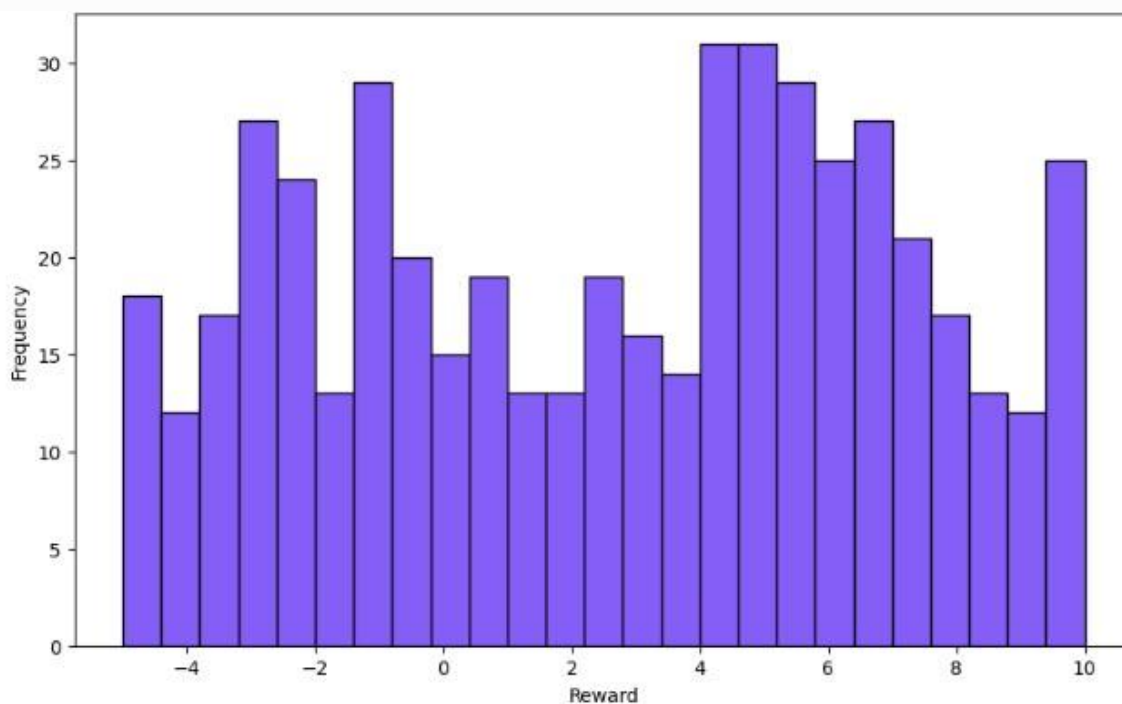


FIG: - 4 DATA SET REWARD DISTRIBUTION

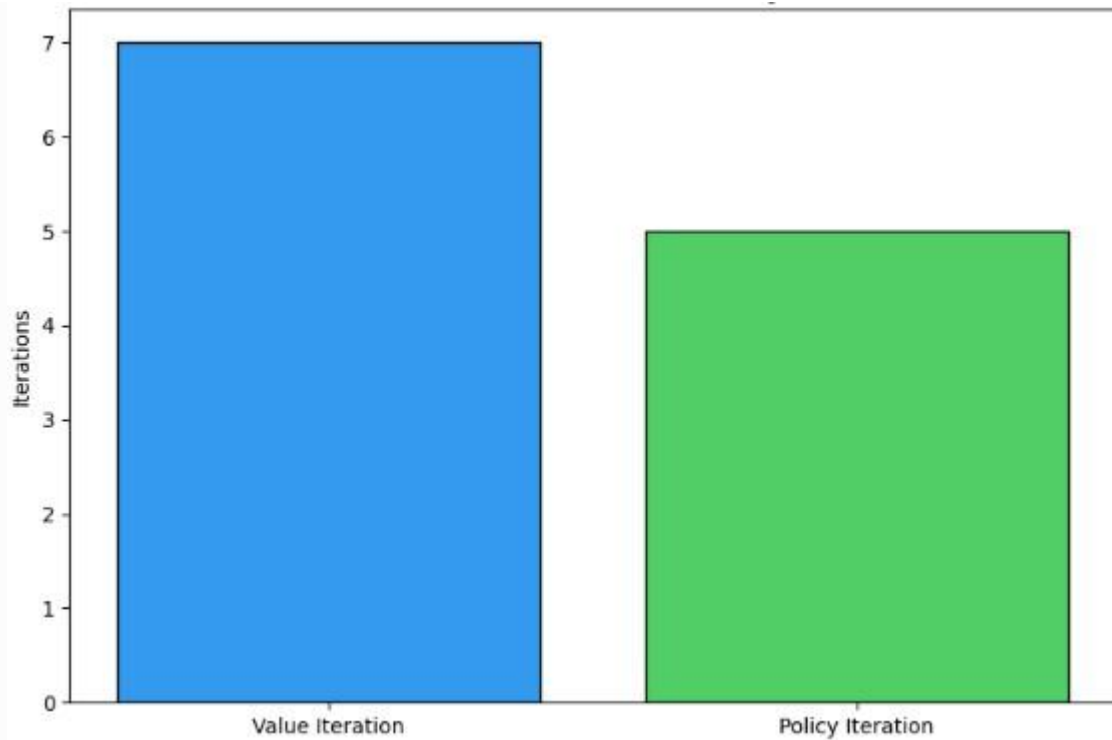


FIG: - 5 VISUALIZATION OF ALGORITHM COMPARISON

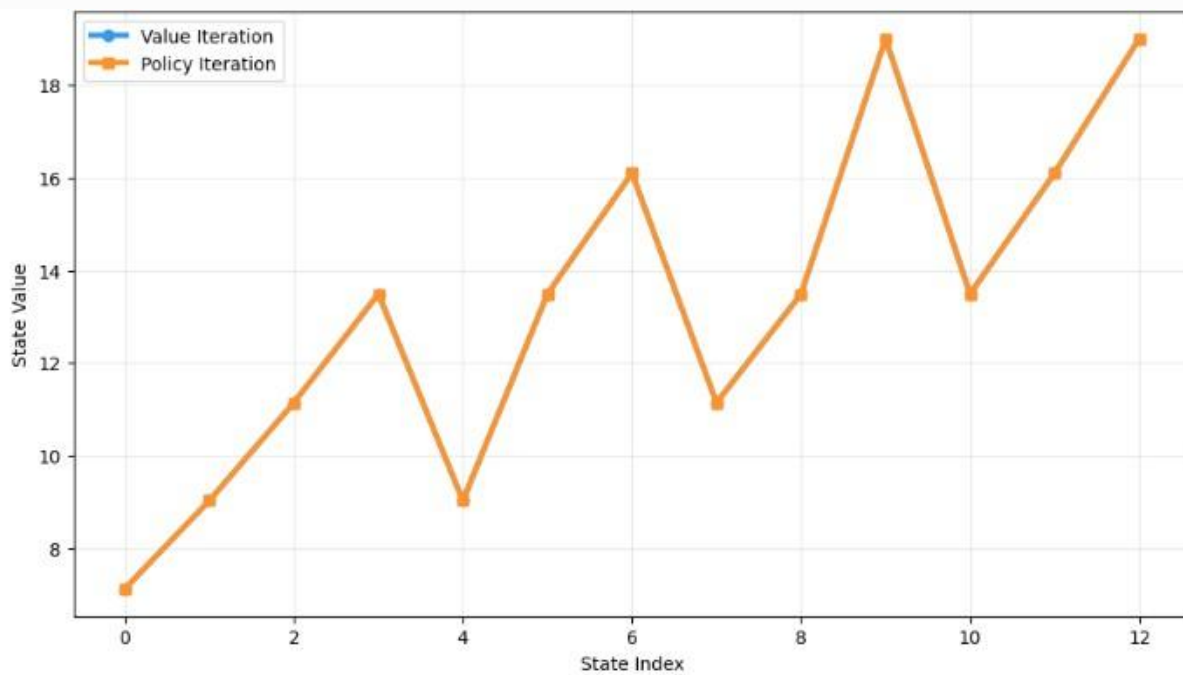


FIG: - 6 STATE VALUE COMPARISON

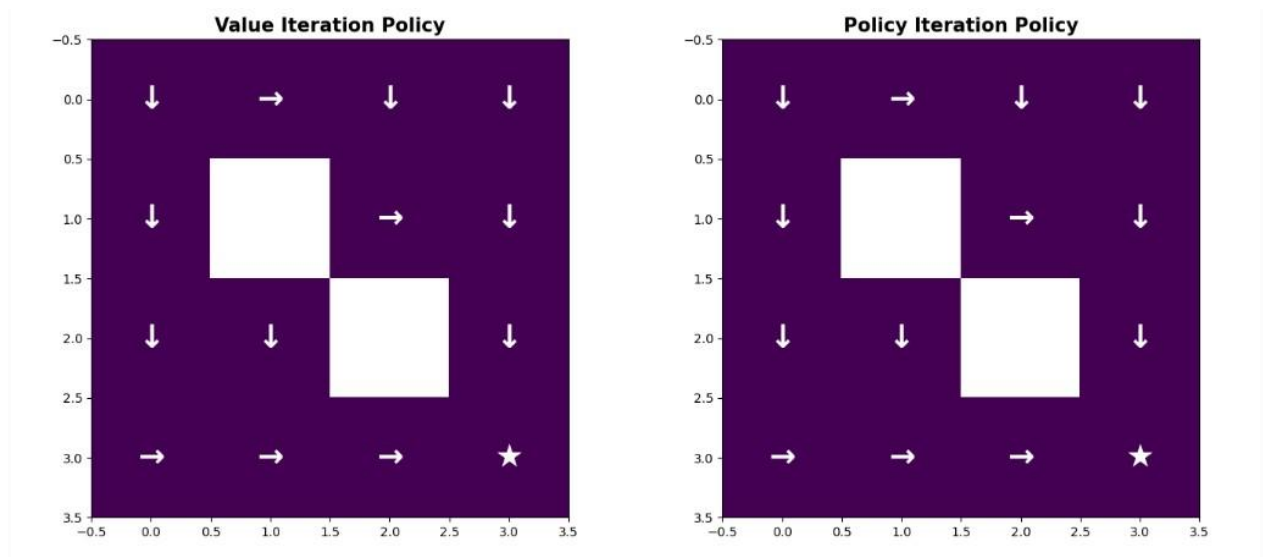


FIG: - 7 FINAL COMPARISON

Result

Both Policy Iteration and Value Iteration were successfully implemented. The optimal value function and optimal policy were obtained, and their performance was compared.